Xinglu **Wang**

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Education

Simon Fraser University — *Burnaby, British Columbia*

Ph.D. in Computing Science — *Sep 2021–Jun 2026*

- Supervised by **Prof. Jian Pei** and **Prof. Jiannan Wang**. GPA: 4.13/4.33.
- Helmut and Hugo Eppich Family Graduate Scholarship, Graduate Fellowships, Special Graduate Entrance Scholarship.

Zhejiang University — *Hangzhou, Zhejiang*

M.S. in Information Engineering — *Sep 2018–Jun 2021*

- Supervised by **Prof. Yingming Li** and **Prof. Zhongfei (Mark) Zhang**. GPA: 90.75/100.

Zhejiang University — *Hangzhou, Zhejiang*

B.Eng. in Information Engineering — *Sep 2014–Jun 2018*

- GPA: 90.21/100 (equivalent to 3.93/4.00); Rank: 7/174.
- **First-Class Scholarship** for Outstanding Students; Meritorious Winner, **Interdisciplinary Contest in Modeling** (ICM).

Experience

Stealth

Building Something New — *2026–Present*

Quantum Technology Recruiting Inc. — *Burnaby, British Columbia*

Researcher (Contract) — *Aug 2026–Nov 2026*

Employed by Quantum Technology Recruiting Inc., assigned to one of its clients.

Post-training RL for Agentic Workloads

- **Long-Horizon Visual-Agent RL.** Developed a Qwen3.5-9B visual agent for long-horizon Pacman control from raw game-frame observations. Designed an **AReaL**-based RL training pipeline with state-dependent *constrained decoding* and maintained trajectory consistency between vLLM rollout and FSDP training. The resulting agent generalizes to unseen ghost-enabled mazes and successfully completes long-horizon gameplay tasks (see technical blog).
- **Search-Agent RL Training.** Diagnosed reward stagnation in a **MiniMax-M2.5** search agent. Built a clean **verl** recipe for a tool-augmented search agent. Fixed MiniMax-specific tool-call parsing and addressed token-boundary drift using **token-in, token-out (TITO)**. Implemented resumable partial rollouts with version/log-probability tracking for efficient asynchronous training. Orchestrated large-scale training across 8 rollout nodes and 16 training nodes. Achieved **one-sample overfitting**, partially validating the training infrastructure, agent workflow, and reward design.
- **Multi-Agent RL Training.** Implemented role-aware GRPO reward normalization for shared-backend Chain-of-Agents MARL in **AReaL**, validated it on mathematical reasoning tasks, and submitted it as AReaL PR #1129.

SFU Data Science Research Group, Simon Fraser University — *Burnaby, British Columbia*

Ph.D. Researcher — *Sep 2021–Jun 2026*

Concurrent role: Associate Researcher Intern, Quantum Technology Recruiting Inc.; assigned to a technology-sector client

Failure Recovery for Large-scale MoE Serving — *Aug 2025–Feb 2026*

- Designed and implemented the core recovery path for disaggregated attention/MoE workers in **ReviveMoE** [1], enabling fast recovery without full instance restarts. Built abstractions for attention and MoE executors and **proxy-based switching** for rapid role transitions.
- Developed a step-level recovery framework enabling safe rollback from unknown intermediate states via **transaction-log-based recovery**, including block-table restoration, request migration, and partial recomputation.

SLO-aware LLM Serving System — *Dec 2024–Aug 2025*

- Designed heuristic **SLO-aware** dispatching and scaling algorithms for LLM serving with heterogeneous latency targets, balancing latency requirements and serving cost. Published on *arXiv 2025* [2].
- Implemented **fast-scaling** mechanisms for disaggregated serving, including pre-started serving processes, device-to-device weight transfer, and stage-aware orchestration.

Disaggregated Large Multimodal Model Serving *Feb 2024–Dec 2024*

- Evaluated an **Encode–Prefill–Decode (EPD)** disaggregated serving system for large multimodal models, analyzing efficiency and limitations across scenarios; co-authored the **ICML 2025** paper [4].
- Analyzed inter-stage coordination, cache layouts, and memory management in EPD serving, including peer-to-peer device communication, **CUDA IPC** memory sharing, and CUDA virtual memory management.

LLM for Mathematical Modeling *Feb 2023–Feb 2024*

- Evaluated prompt engineering and supervised fine-tuning (SFT) for LLM-based mathematical modeling across optimization problems of varying complexity; wrote substantial portions of an **EJOR 2026** survey [3] covering how AI can enhance multiple stages of the optimization pipeline.
- Proposed graph-edit-distance evaluation for linear-programming word problems that recognizes mathematically equivalent formulations beyond exact match, including differences in constraint ordering and variable naming; published at **LREC-COLING 2024** [6].

Contribution Valuation for Vertical Federated Learning *Feb 2023–Feb 2024*

- Co-developed **VerFedSV**, a Shapley-value metric that reuses timestamped VFL embeddings from one training trajectory to avoid coalition-wise retraining while preserving four fairness axioms.
- For synchronous VFL, reconstructed missing full-dataset embeddings from mini-batch observations via low-rank matrix completion; for asynchronous VFL, directly computed valuations that reflect clients' compute and upload capacity, with no extra communication; published at **ICLR 2024** [7].

Smart Initial Basis Selection for Linear Programs *Feb 2022–Feb 2023*

- Proposed a graph neural network approach to predict **close-to-optimal** initial bases for linear programming solvers.
- Combined model prediction with post-processing steps to generate valid bases for simplex warm starts.
- Demonstrated substantial speedup over rule-based baselines across multiple datasets and solvers; published at ICML 2023 [8].

Data Science & Engineering Research Center, ZJU *Hangzhou, Zhejiang*

Master and Undergraduate Theses

Harmonized Multi-exit Learning *Dec 2019–Aug 2021*

- Studied multi-exit learning for adaptive inference, addressing optimization interference between exits.
- Proposed gradient deconfliction training using gradient projection to resolve exit conflicts; published at **IEEE ICIP 2020** [13].
- Designed a meta-learning-based harmonized weighting scheme for dense teacher-student distillation across exits; accepted at **AAAI 2021** [11].
- Evaluated on large-scale ImageNet using Cloud TPU and reported a bug in **pytorch/xla**.

Person Re-identification *Oct 2017–Jun 2018*

- Analyzed data sampling, feature extraction, loss design, and post-processing for person re-identification (see technical blog).
- Proposed a Cython module in **deep-person-re-id** to accelerate evaluation by **20**×, and fixed a ResNet depth bug in **pytorch-classification** to enable fair model comparison.

Publications

[1] Li, H., **Wang, X.**, et al. 2026. ReviveMoE: Fast recovery for hardware failures in large-scale MoE LLM inference deployments. arXiv preprint arXiv:2602.21140.

[2] Yousefijamarani, Z., **Wang, X.**, et al. 2025. HFX: Joint Design of Algorithms and Systems for Multi-SLO Serving and Fast Scaling. arXiv preprint arXiv:2508.15919.

[3] Fan, Z., Ghaddar, B., **Wang, X.**, et al. 2026. Artificial intelligence for optimization: Unleashing the potential of parameter generation, model formulation, and solution methods. *European Journal of Operational Research*, 332(1), 1–30.

[4] Singh, G., **Wang, X.**, et al. 2025. Efficiently serving large multimodal models using EPD disaggregation. In *ICML*.

[5] Heisler, M., Yousefijamarani, Z., **Wang, X.**, et al. 2025. LLM inference scheduling: A survey of techniques, frameworks, and trade-offs. *TechRxiv Preprint*. DOI: 10.36227/techrxiv.176238087.79673350.

[6] Xing, L., **Wang, X.**, et al. 2024. Towards human-aligned evaluation for linear programming word problems. In *LREC-COLING* (pp. 16550–16556).

[7] Fan, Z., Huang, F., **Wang, X.**, et al. 2024. Fair and efficient contribution valuation for vertical federated learning. In *ICLR*.

[8] Fan, Z., **Wang, X.**, et al. 2023. Smart initial basis selection for linear programs. In *ICML* (pp. 9650–9664).

[9] Gholami, M., Akbari, M., **Wang, X.**, et al. 2023. ETran: Energy-based transferability estimation. In *ICCV* (pp. 18613–18622).

[10] Qiao, C., Xiang, Z., **Wang, X.**, et al. 2023. Objects matter: Learning object relation graph for robust absolute pose regression. *Neurocomputing*, 521, 11–26.

[11] **Wang, X.**, and Li, Y. 2021. Harmonized dense knowledge distillation training for multi-exit architectures. In *AAAI*, 35(11), 10218–10226.

[12] Ouyang, S., **Wang, X.**, et al. 2021. Pseudo-label generation-evaluation framework for cross-domain weakly supervised object detection. In *ICIP* (pp. 724–728).

[13] **Wang, X.**, and Li, Y. 2020. Gradient deconfliction-based training for multi-exit architectures. In *ICIP* (pp. 1866–1870).